

Introduction to Social Data Analytics: Week 10 Part 2

Today: Conclusion

- ▶ Review each week at a high level
 - ▶ This will not include a review of every command/function you need to know, but will point out some of the more important ones
- ▶ Practice Questions

Quick Overview of Topics

- ▶ We studied a lot of different papers this year
- ▶ Let's briefly review what we learned from each topic

Week 1: College Instruction and Financial Incentives

- ▶ Question: Is it possible financial incentives to instructors might improve teaching?
- ▶ In Brownback and Sadoff (2020) the authors run an experiment. At a community college in Indiana (Ivy Tech) some instructors (the treatment group) received \$50 for each student that passes the test, while the control group did not receive such an incentive
- ▶ Show that pay-for-performance very effective in this setting

Week 1: Important lessons

- ▶ 1. What is the motivation behind experiments? What do they allow us to infer relative to observation data
- ▶ 2. Excel functions will not be on the final

Week 2: Intergenerational Mobility Rates by College

- ▶ If you are born from low-income parents, what is the chance you move up the income distribution?
- ▶ Higher education is often viewed as a path for upward mobility
 - ▶ Differences in quality across colleges and rising costs of college may mean lower access/success for lower-income students
- ▶ Researchers at **Opportunity Insights** allowed us to understand variation across colleges in the extent that they promote intergenerational mobility
 - ▶ Found large variation, with some schools essentially having zero mobility

Week 2: Important lessons

- ▶ Stata lessons
 - ▶ How do we generate new variables, construct summary statistics, and plot histograms
- ▶ Conceptual lessons
 - ▶ What is a binary variable and how do we generate them from logical statements
 - ▶ What is a histogram and when is it useful

Week 3: Stanford Open Policing Project

- ▶ The most common way civilians interact with police are through traffic stops (there are about 50,000 a day)
- ▶ Detecting and eliminating racial discrimination in police behavior is an important policy goal, but can be difficult to detect without data
- ▶ **The Stanford Open Policing Project** aggregated data on police stops across states and cities so that researchers and journalists and policymakers can improve interactions between police and the public
- ▶ We implemented the veil of darkness test to study discrimination in San Diego

Week 3: Important lessons

- ▶ Stata lessons
 - ▶ What are the differences between append, merge and collapse. When is each useful?
- ▶ Conceptual lessons
 - ▶ Often we need to restructure our data to accomplish our analysis. Important to understand the structure of the data so that you can get it into the proper format
 - ▶ This is something we applied throughout the course (especially week 8)

Week 4: Disrupting Education using Technology

- ▶ If curriculum is beyond a student's level of understanding, they may learn very little, even if attendance is high
- ▶ Question: can technology that aims to “Teach at the Right Level” be used to improve learning outcomes
- ▶ We studied the results of an experiment in India that randomized access to centers with technology that aimed to teach at the right level

Week 4: Important lessons

- ▶ Stata lessons
 - ▶ Using `reg` to estimate linear regressions
- ▶ Conceptual lessons
 - ▶ How do we interpret the slope coefficient and intercept
 - ▶ How do we interpret regression with continuous vs. binary variables

Week 6: Discrimination in the Labor Market

- ▶ Pervasive evidence of inequality in the labor market
 - ▶ Black workers are about twice as likely as white workers to be unemployed (Council of Economic Advisors, 1998)
- ▶ However, often difficult to isolate discrimination from employers as source of a given disparity
- ▶ We studied the results of Bertrand and Mullainathan (2004) that randomized names across fictitious resumes to identify discrimination

Week 6: Important Lessons

- ▶ R lessons:
 - ▶ Creating objects and applying functions to vectors
 - ▶ subsetting data using indexing
 - ▶ Changing the working directory and loading in data
- ▶ Conceptual lessons
 - ▶ What are objects in R?
 - ▶ How do logical operations work?

Week 7: The Rug Rat Race

- ▶ Amount of time spent on childcare rose dramatically in 1990s
- ▶ Mothers in particular, college-educated parents in particular
- ▶ Why? Puzzling because these parents also had more to gain in the labor market
- ▶ Ramey and Ramey (2010) use time diaries to argue that this rise in childcare time is driven by the rise in competitiveness of colleges

Week 7: Important Lessons

- ▶ R lessons:
 - ▶ For loops
 - ▶ The tidyverse package: `arrange`, `filter`, `mutate`, `select`, `group_by` and `summarize`
- ▶ Conceptual lessons:
 - ▶ We often need to change structure of the data (data wrangling) to accomplish some task. This week introduced tidyverse which we often use to accomplish this in other weeks as well

Week 8: Automated Air Pollution Monitoring in China

- ▶ If environmental policy goals are to be met, they need to be measured accurately
- ▶ Greenstone et al. (2022) find evidence of local manipulation of pollution numbers
- ▶ After automatic monitoring went into place, the amount of pollution spiked
 - ▶ This wasn't a real spike in pollution, but reflects the fact that reports were artificially low before the automated monitoring

Week 8: Important lessons

- ▶ R lessons
 - ▶ Base R plots: `plot()`, `hist()`, `boxplot`
 - ▶ `ggplot2`: an external package for generating highly customizable figures
- ▶ Conceptual lessons
 - ▶ What makes a good figure (i.e. clear labels, proper titles, easily interpretable)
 - ▶ How do we interpret the output of all the figures discussed in both Base R and `ggplot2`

Week 9: Voting Applications

- ▶ Application 1: A poorly designed ballot in a single Florida county may have changed the 2000 presidential election!
- ▶ Application 2: Why don't individuals vote in primaries?
 - ▶ It may be lack of mobilization
 - ▶ It may be lack of information about primaries
 - ▶ Hill and Kousser (2016) implemented a massive experiment to understand what motivates individuals to vote in primaries

Week 9: Important lessons

- ▶ R lessons
 - ▶ Using `lm()` to estimate linear regressions
- ▶ Conceptual lessons
 - ▶ How to interpret the output of a linear regression

Week 10: Unconditional Cash Transfers

- ▶ How to alleviate poverty and improve welfare is one of the most important questions development economists tackle
- ▶ Haushofer and Shapiro (2016) studies the impact of unconditional cash transfers in Kenya on the economic and psychological well-being of recipients

Week 10: Important Lessons

- ▶ R lessons:
 - ▶ How do we write our own function in R
- ▶ Conceptual lessons
 - ▶ Why is it useful to write our own functions?

Practice Questions

Concluding Thoughts

- ▶ Analyzing data can be a powerful tool in business, academics and policy
- ▶ There is so much data available today that it is likely to become even more important in the future
- ▶ I hope this class has been a first step for you in understanding how to interpret and analyze data
- ▶ Please keep in touch in the future!